
An Exact Conditional Null for Correlated Failure in Open-Model Ecosystems, and a Route to Anytime-Valid Monitoring

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Abstract

1 Measuring correlated failure between language models requires a null hypothesis
2 for “independent enough,” and prior work shows the conclusion is highly sensitive
3 to which null is chosen, leaving the choice apparently subjective. We show one null
4 is not a choice: the row and column margins of a model-by-item outcome matrix
5 are the jointly sufficient statistics of the Rasch family, so the uniform distribution
6 over margin-matched matrices is the unique exact, fit-free conditional null for
7 that family. At this null, on 1,228–1,362 open models across five benchmarks,
8 the commonly reported mean-level “excess” co-failure is a deterministic function
9 of item difficulty and vanishes identically, while a genuine second-order residual
10 correlation of $2.8\text{--}11.8\times$ its null survives, is robust to removing near-duplicate
11 models, and concentrates in a handful of latent dimensions. Our test is currently
12 a fixed-sample exact randomization test. We argue the same conditional null is
13 the natural fixed-sample building block for an anytime-valid monitor of correlated
14 failure as an open-model ecosystem grows, and sketch the construction as an open
15 problem for this workshop.

1 Motivation

17 If many deployed models share blind spots, an ensemble, a panel of model judges, or a market of
18 competing screening vendors supplies far less independent evidence than its headcount suggests.
19 Measuring this requires comparing the rate at which models fail the same items against a baseline
20 in which failures are independent. Jo et al. (2026) show this comparison is highly sensitive to the
21 assumed baseline: conditioning on item difficulty attenuates residual correlations, “some even flipping
22 from strongly positive to slightly negative,” and they prove that a sufficiently expressive null can
23 explain away any observed dependence. Their stated conclusion is that the choice of null is subjective.
24 This paper argues the ladder of nulls has a canonical rung, reports what the exact answer is there, and
25 points at what this workshop’s machinery could add to it.

26 2 The canonical null

27 Let $F \in \{0, 1\}^{N \times M}$ record $F_{im} = 1$ iff model i fails item m , with row sums r_i and column sums
28 c_m . The Rasch model $\Pr(F_{im} = 1) = \sigma(\alpha_i + \beta_m)$ has $\{r_i\}$ and $\{c_m\}$ as jointly sufficient statistics.
29 Conditioning on both therefore eliminates α and β exactly: the conditional law of F given its margins
30 is uniform on the set of margin-matched binary matrices, for every parameter value (Rasch, 1960).
31 That set is the ecologists’ fixed-fixed null, and we sample it with curveball (Strona et al., 2014), whose
32 convergence to uniform on the fibre is proved by Carstens (2015) and which we verify directly by
33 enumeration on four small fibres. Unlike a fitted item-response null, this null requires no optimizer, no

Benchmark	N	$N_{\text{eff obs.}}$	$N_{\text{eff null}}$	ratio	rms excess ($\times$ null)
ARC-Challenge	1362	23.6	449.3 ± 3.4	0.052	$5.5\times$
Winogrande	1361	20.1	493.8 ± 3.0	0.041	$4.6\times$
TruthfulQA	1334	17.4	309.7 ± 2.4	0.056	$4.1\times$
GSM8K	1228	118.6	553.1 ± 2.2	0.214	$2.8\times$
HellaSwag	1362	27.3	958.0 ± 3.2	0.028	$11.8\times$

Table 1: Effective independent models after exact-margin conditioning, all five benchmarks. The null column is the mean and SD of 400 curveball replicates with both margins exactly preserved.

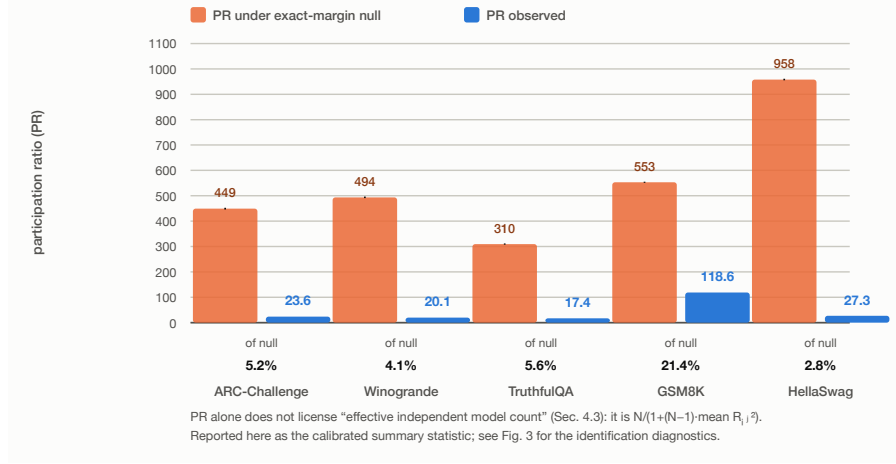


Figure 1: Observed participation ratio against its exact-margin null, all five benchmarks. Reported here as a calibrated summary statistic, not as a literal count of independent models (Section 4’s test cannot distinguish one weak factor from tight clusters).

34 penalty, and no identification convention — it is the strongest ability-and-difficulty control obtainable
35 without fitting anything, and it is unique in that respect.

36 **Lemma 1 (Mean co-failure depends only on item margins)** $O = \frac{1}{MN(N-1)} \sum_m (c_m^2 - c_m)$.

37 Consequently any randomization preserving the item margins leaves the mean pairwise co-failure
38 rate O invariant with exactly zero variance: the reported “excess” co-failure at this null is identically
39 zero by construction, not empirically small. This degeneracy is independently known in ecology
40 (Schluter’s V-ratio), psychometrics (Cronbach’s α), and inter-rater agreement (Fleiss), none of which
41 is cited in the model-evaluation literature that reports it; we state the identities and attribute them in
42 the full paper.

43 3 What survives at the exact null

44 Table 1 reports the participation ratio of the conditioned residual-correlation spectrum against its
45 exact-margin null, on the archived Open LLM Leaderboard (1,228–1,362 models, five benchmarks,
46 harvested at zero inference cost by reading only the metric column of each result file). The observed
47 value sits at 2.8–21.4% of the null. Uncalibrated, the same statistic reads 1.5–3.6 on every benchmark
48 — a number driven almost entirely by the single shared item-difficulty eigenvalue the conditioning
49 removes, and easily misread as “near-total redundancy” before calibration. We show algebraically
50 that this statistic cannot separate one weak global factor from a partition into tight clusters, and we use
51 spectral diagnostics instead: on ARC, the leading eigenvalue explains only 54% of squared residual
52 mass (a single shared factor would give $\approx 99\%$), and six eigenvalues exceed the null’s spectral edge.
53 The result is robust to removing near-duplicate models (on ARC, discarding 900 of 1,362 near-clones
54 moves the ratio from 23.6 only to 29.0) and to a $2.8\times$ -larger replication population collected six
55 weeks later, where every ratio moves by at most 0.010.

A pre-registered dispersion statistic (the variance of pairwise co-failure) was our primary planned test and it failed its own kill condition, coming out inconsistent in sign across benchmarks; we report it as refuted rather than reinterpreted. What survives is the second-order spectral structure above, not the first statistic we planned to report — a distinction we think matters for how confirmatory claims about model diversity should be validated in this literature.

4 Toward an anytime-valid version

Everything above is a fixed- n exact test: for a fixed population of N models, curveball gives Monte Carlo draws from the exact conditional distribution, and the observed statistic is compared against them once. The population it is computed on is not actually fixed — the Open LLM Leaderboard and its successors add new open models continuously, and a natural downstream use of this test is a standing monitor: “has correlated failure in the open ecosystem gotten worse since last quarter,” checked repeatedly as new submissions arrive, without the type-I error inflation that repeated testing on a growing sample otherwise incurs. That is exactly the anytime-valid inference problem this workshop’s community formalizes with e-values and test martingales (Shafer, 2021; Vovk & Wang, 2021; Ramdas et al., 2023; Grünwald et al., 2024).

We see two routes and neither is worked out here. First, one could construct a test martingale directly from the exact conditional law: since the conditional distribution of F given its margins is uniform on a fully specified finite set for every model added, each new model’s observed row is itself a draw whose conditional distribution under the null is known exactly (not merely asymptotically), which is the situation in which e-value construction from a likelihood ratio against a composite alternative is most tractable. Second, since our headline statistic is already a Monte Carlo p -value against an exact reference distribution, one could wrap it in a merging or calibration scheme that converts a sequence of such p -values into an e-process, trading some power for the anytime-validity guarantee. The obstacle in both cases is that new models are not exchangeable arrivals in the usual online-testing sense: they are added by submitters who observe the current leaderboard, so any construction has to either condition away that selection effect or treat it explicitly — a version of the same population-dependence objection Jo et al. (2026) raise about a fixed null, now posed sequentially. We flag this as the concrete open problem we would bring to this workshop, not as a result: the exact conditional null is, we think, the right fixed-sample primitive to build an anytime-valid ecosystem monitor on top of, but we have not yet built one.

5 Limitations

Claims cover the evaluated open-model ecosystem on multiple-choice-style benchmarks, not deployed proprietary systems. Our conditioning model is unidimensional (Rasch); a misspecification check shows two-parameter discrimination heterogeneity alone cannot reproduce the observed effect size, but a low-dimensional multi-ability structure can reproduce much of it, so we cannot fully separate genuine behavioral concentration from an underfit null family. All code, data, and the pre-registration (with dated amendments) are archived and linked in the full paper.

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